

REFERENCES

1. de Berker DAR, Rendall JR. Retronychia-proximal ingrowing nail. *J Eur Acad Dermatol Venereol* 1999;12:S126.
2. Baumgartner M, Haneke E. Retronychia: diagnosis and treatment. *Dermatol Surg* 2010;36:1610-4.
3. de Berker DA, Richert B, Duhard E, Piraccini BM, André J, Baran R. Retronychia: proximal ingrowing of the nail plate. *J Am Acad Dermatol* 2008;58:978-83.
4. Chiheb S, Richert B, Belyamani S, Benchikhi H. Ingrown nail: a new cause of chronic perionyxis. *Ann Dermatol Venereol* 2010;137:645-7.
5. Dahdah MJ, Kibbi AG, Ghosn S. Retronychia: report of two cases. *J Am Acad Dermatol* 2008;58:1051-3.

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Pruritus severity in patients with psoriasis is not correlated with psoriasis disease severity

To the Editor: Pruritus has been reported in the majority of psoriatic patients, with 85% affected, and 77% having pruritus daily.^{1,2} In a recent study, 80% of psoriatics reported that pruritus was the most important and severe symptom.³ Pruritus has also been found to have a significant impact on the quality of life of psoriatics, with regards to falling asleep, and poorer psychosocial well-being, with patients found to exhibit feelings of stigmatization and depressive symptoms.^{1,3,4} There is significant unmet need for effective treatment of the pruritus of psoriatics. In all, 45% of patients reported that no treatment relieved their pruritus, and only 8% of patients on topical corticosteroids found their pruritus was reduced.¹ Despite this, no therapies that specifically treat pruritus exist. Indeed, few clinical trials have assessed pruritus, and it is not part of any validated psoriasis scoring indices such as the Psoriasis Area Severity Index (PASI) or Investigator Global Assessment.

We report baseline data of 157 psoriatics enrolling for a clinical trial (EudraCT number: 2011-004640-21). These patients had mild to moderate psoriasis according to Investigator Global Assessment and a mean modified PASI (mPASI) score of 8.9. We found nearly all patients (97.5%) had pruritus; 68.8% had at least moderate pruritus (visual analog scale [VAS] score ≥ 40 mm) and 33.8% severe pruritus (VAS score ≥ 70 mm). This agrees with reports in the literature that suggest that 70% of psoriatics have at least moderate pruritus.¹

Conflicting reports exist in the literature as to whether pruritus is associated with disease severity.^{2,4,5} We thus explored whether there was a relationship between pruritus, measured by VAS, and psoriasis disease severity, measured by mPASI. mPASI primarily measures inflammatory-mediated aspects of psoriasis and does not directly measure pruritus. The lack of measurement of pruritus in the

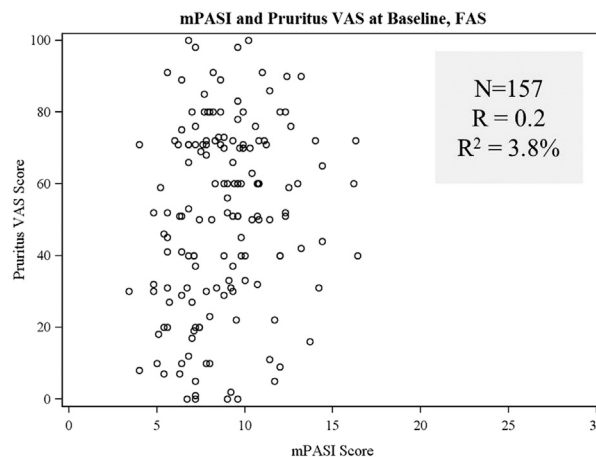


Fig 1. Lack of correlation between modified Psoriasis Area and Severity Index (mPASI) and pruritus visual analog scale (VAS) scores of study patients with psoriasis. Pruritus VAS scores, measuring pruritus severity, and mPASI scores, measuring psoriasis disease severity, were recorded in 157 patients at baseline enrolling for a clinical trial. The Psoriasis Area and Severity Index (PASI) score was modified as the scalp and face were not assessed. In all, 160 patients were enrolled in the trial, but 3 patients were excluded from the correlation analysis as they did not have properly recorded VAS or mPASI scores. The correlation between VAS and mPASI scores in these patients was measured through the coefficient of determination (R^2). A lack of correlation between VAS and mPASI scores was found. The R^2 value was 3.8%, meaning that only 3.8% of the variability in pruritus severity could be explained by its relationship to psoriasis disease severity. FAS, Full analysis set.

PASI scale should reduce bias when comparing the 2 indices. Fig 1 shows a scatterplot of pruritus severity (VAS) against psoriasis disease severity (mPASI) for the patients in the study. We found no evidence of any clinically meaningful correlation between pruritus severity and psoriasis disease severity. The coefficient of determination (R^2) value was 3.8%, meaning that only 3.8% of the variability in pruritus severity can be explained by its relationship to disease severity as measured by mPASI.

As mPASI measures inflammatory-mediated aspects of psoriasis, this lack of correlation suggests that treating pruritus in patients with psoriasis with anti-inflammatory agents may be a suboptimal approach. Current treatments for patients with mild-moderate psoriasis such as topical corticosteroids (glucocorticoids) and vitamin D analogs focus on treating the inflammatory component of psoriasis, with the intent that this will also treat any pruritus. Surprisingly, given the importance of pruritus to patients with psoriasis, the impact of these agents on

pruritus in psoriasis has not been effectively studied. The lack of correlation between pruritus severity and psoriasis disease severity that we have observed, coupled with the high prevalence of pruritus and unmet need, warrants greater attention to pruritus by clinicians and researchers to develop targeted antipruritic therapies for patients with psoriasis and other dermatologic conditions.

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REFERENCES

1. Prignano F, Ricceri F, Pescitelli L, Lotti T. Itch in psoriasis: epidemiology, clinical aspects and treatment options. *Clin Cosmet Invest Dermatol* 2009;2:9-13.
2. Yosipovitch G, Goon A, Wee J, Chan YW, Goh CL. The prevalence and clinical characteristics of pruritus among patients with extensive psoriasis. *Br J Dermatol* 2000;143:969-73.
3. Globe D, Bayliss MS, Harrison DJ. The impact of itch symptoms in psoriasis: results from physician interviews and patient focus groups. *Health Qual Life Outcomes* 2009;7:62.
4. Reich A, Hrehorow E, Szepietowski JC. Pruritus is an important factor negatively influencing the well-being of psoriatic patients. *Acta Derm Venereol* 2010;90:257-63.
5. Szepietowski JC, Reich A, Wiśnicka B. Itching in patients suffering from psoriasis. *Acta Dermatovenereol Croat* 2002;10:221-6.

Treatment of erythropoietic protoporphyria with the oral sorbent colestipol: A proof-of-concept clinical trial

To the Editor: Erythropoietic protoporphyria (EPP) is characterized by cutaneous photosensitivity and, rarely, potentially lethal liver dysfunction. The accumulation of free protoporphyrin IX in tissues and organs of patients with EPP results from mutations in genes coding for enzymes involved in heme biosynthesis.¹ Hepatocyte protoporphyrin accumulation, the cause of hepatic failure, is augmented by both its transfer from blood and the enterohepatic recirculation of porphyrins after they leave the biliary tract.² Several treatments, often marginally effective, are available for the cutaneous manifestations.³ No medical therapies prevent the progression of hepatic dysfunction.

In vitro, cholestyramine and activated charcoals can irreversibly bind porphyrins, including protoporphyrin.⁴ Their irreversible binding of ileum-based protoporphyrin, preventing its enterohepatic reabsorption, may lead to decreased accumulation of protoporphyrin and be therapeutic in both skin and liver. This study addresses the hypothesis that a useful sorbent, colestipol, effectively diminishes the cutaneous and hepatic manifestations of EPP.

This study, approved by Partners HealthCare Investigational Review Board, included 3 adult women with EPP and cutaneous but no hepatic manifestations. After a period of no therapy (during late winter and spring), each subject received oral colestipol at 2 g twice daily (during summer and early autumn). In both phases, subjects used a stopwatch to determine the time to the onset of discomfort with sun exposure and completed a standardized questionnaire, addressing the affected areas, and the nature and severity of manifestations. Blood samples were analyzed for liver function and

Table I. Sun tolerance with colestipol 2 g twice daily

Days of treatment	Onset of sensitivity to sun*					
	Subject 1		Subject 2		Subject 3	
	Min	Fold ↑†	Min	Fold ↑†	Min	Fold ↑†
0	8	—	1.6	—	9.3	—
60-65	74	9.3	3.3	2.1	120	13
125-127	190	24	2.5	1.6	∞‡	∞‡
157-163	220	28	7.1	4.4	∞‡	∞‡

*Determined by each subject recording the onset of skin discomfort with a stopwatch.

†Colestipol-mediated increase in sun tolerance relative to baseline tolerance. The hazard per day ratio using the Cox proportional hazards model = 0.96; *P* = .018.

‡No development of skin sensitivity.